2) **Logistic Regression**

Probability vs Odds

Q) A guy went fishing 5 times a week, caught fish 2 times failed to catch 3 times. What is the probability and odds of getting a fish for dinner ?

A)

$$\text{Probability} = \frac{\text{chances for}}{\text{total chances}} = \frac{2}{5}$$

$$\text{Odds} = \frac{\text{chances for}}{\text{chances against}} = \frac{2}{3}$$

$$= \frac{P}{1-P}$$

argument

$$\log_b(n) = x \iff b^x = n$$

base, exponent

Age	Salary	Loan
23	1,50,000	Yes
40	15,000	No

$y = mx + c$

$y = m_1 y_1 + m_2 y_2 + c$

$y = m_1(\text{age}) + m_2(\text{salary}) + c$

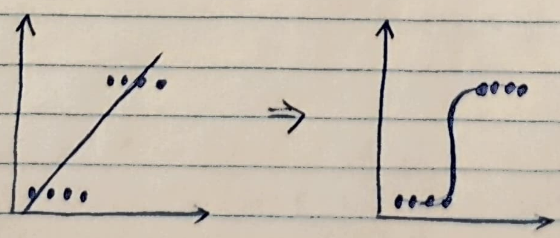
$$\log\left(\frac{P(x)}{1-P(x)}\right) = mx + c$$

Sigmoid Eq^n

$$\boxed{g(x) = \frac{1}{1+e^{-z}}} \Rightarrow (0,1)$$

$$g(x) = \frac{1}{1+e^{-(m_1(\text{age}) + m_2(\text{salary}) + c)}} = (0,1)$$

m_1, m_2 → Maximum likelihood estimation



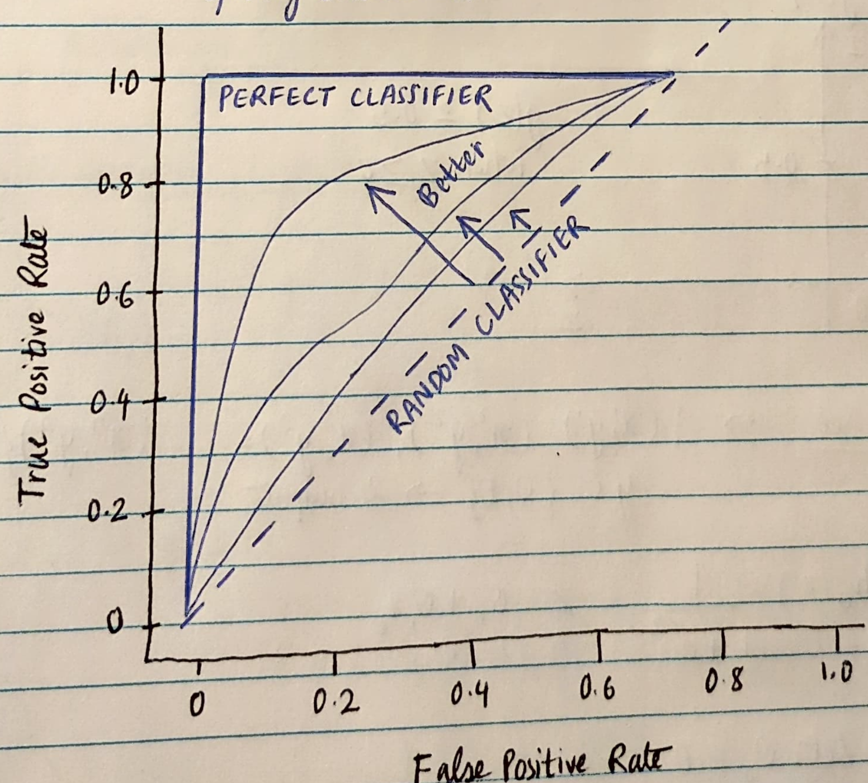
Confusion matrix

	Predicted Positive	Predicted Negative	True +ve Rate
Actual Positive	TP True positive	FN False Negative	Sensitivity / Recall $\frac{TP}{TP+FN}$
Actual Negative	FP False Positive	TN True Negative	Specificity $\frac{TN}{TN+FP}$
	Precision $\frac{TP}{TP+FP}$	Negative Prediction value $\frac{TN}{TN+FN}$	Accuracy $\frac{TP+TN}{TP+TN+FP+FN}$

Misclassification Rate $\rightarrow \dfrac{FN+FP}{TP+TN+FN+FP}$

F1 score $\rightarrow \dfrac{2 * Precision * Recall}{Precision + Recall}$

ROC curve (Receiver ~~under~~ curve)
operating characteristics

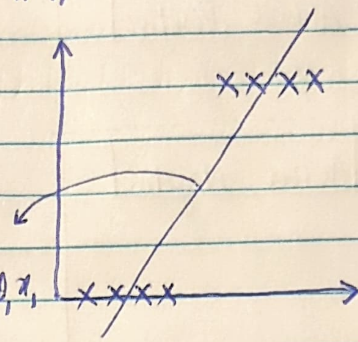


Decision Boundary Logistic Regression

Krish
Naik

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$

$$h_\theta(x) = \theta^T x$$



$$h_\theta(x) = \theta_0 + \theta_1 x_1$$

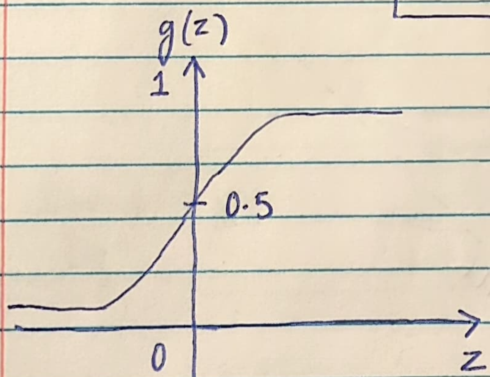
$$h_\theta(x) = g(\theta_0 + \theta_1 x_1)$$

Let $z = \theta_0 + \theta_1 x$

$$h_\theta(x) = g(z)$$

Sigmoid / Logistic function

$$\boxed{h_\theta(x) = \frac{1}{1+e^{-z}} \qquad h_\theta(x) = \frac{1}{1+e^{-(\theta_0+\theta_1 x)}}}$$



$g(z) \geq 0.5$
when $z \geq 0$

Taining Set $\rightarrow \{(x^1, y^1), (x^2, y^2), (x^3, y^3) \dots (x^n, y^n)\}$

$y \in \{0, 1\} \rightarrow$ 2 outputs

$$h_\theta(z) = \frac{1}{1+e^{-z}} \qquad z = \theta_0 + \theta_1 x_1$$

Suppose Let $\theta_0 = 0$

then $z = \theta_1 x_1$

Cost function

Linear Regression $J(\theta_1) = \frac{1}{m} \sum_{i=1}^{m} \frac{1}{2} (h_\theta(x^i) - y^i)^2$

Logistic Regression $h_\theta(x) = \frac{1}{1+e^{-(\theta_1 x)}}$

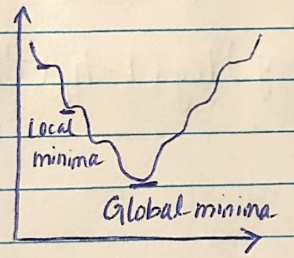
Cost function $= \frac{1}{2}(h_\theta(x^i) - y^i)^2$ } We cannot use this cost function for Logistic Regression

$h_\theta(x) = \underbrace{\frac{1}{1+e^{-(\theta_1 x)}}}$

This is a non-convex function

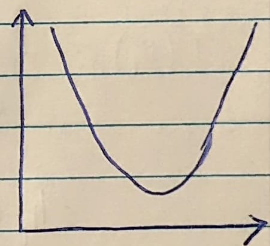
** Gradient Descent

Non-Convex function



if local minima is there then we cannot reach global minima

Convex - function



Logistic Regression Cost function

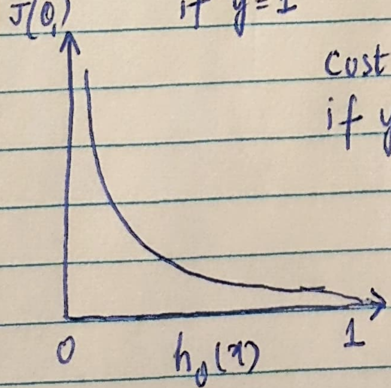
$h_\theta(x) = \frac{1}{1+e^{-(\theta_1 x)}}$

$$J(\theta_1) = \begin{cases} -\log(h_\theta(x^i)) & y=1 \\ -\log(1-h_\theta(x^i)) & y=0 \end{cases}$$

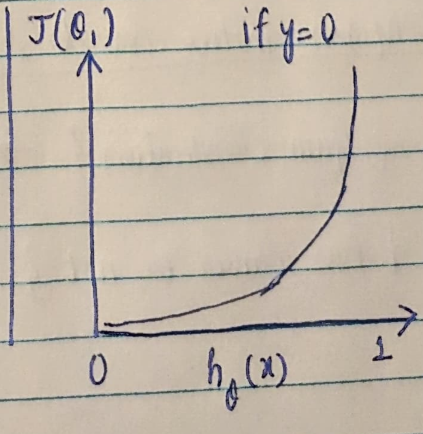
if $y=1$

Cost $= 0$

if $y=1$, $h_\theta(x)=1$



if $y=0$



if we combine these both scenarios then

Cost function

$$(h_\theta(x^i), y) = -y \log(h_\theta(x^i)) - (1-y)\log(1-h_\theta(x))$$

$\Downarrow$

if $y=1$

$$\text{Cost}(h_\theta(x^i), y) = -\log(h_\theta(x^i))$$

if $y=0$

$$\text{cost}(h_\theta(x^i), y) = -\log(1-h_\theta(x^i))$$

$$J(\theta_1) = +\frac{1}{2m}\sum_{i=1}^{m}\left(-y^i\log(h_\theta(x^i)) - (1-y)\log(1-h_\theta(x))\right)$$

Cost function

$$J(\theta_1) = -\frac{1}{2m}\sum_{i=1}^{m}\left(y^i\log(h_\theta(x^i)) + (1-y^i)\log(1-h_\theta(x^i))\right)$$

$$h_\theta(x^i) = \frac{1}{1-e^{-\theta_1 x}}$$

repeat until convergence

{

$$\theta_j := \theta_j - \alpha\frac{\partial}{\theta_j}(J(\theta_1))$$

}

Confusion matrix examples & usage

{ spam classification } → Precision

{ has cancer or not } → Recall

{ Tomorrow stock market is going to crash } POV → for company → Precision
→ for customers → Recall

So combining both the scenarios we use something called as F score

F - Beta $= (1+\beta^2) \dfrac{\text{Precision} \times \text{Recall}}{\beta^2 \times (\text{Precision} + \text{Recall})}$

$\beta = 1$ (F1-score)
→ when FP & FN are important we select β as 1

$$= \frac{(1+1)\ \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2\ \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Harmonic mean $= \dfrac{2xy}{x+y}$ (similar type)

$\beta = 0.5$ (F0.5-score)
→ when FP is more important than FN

$$= \frac{(1+(0.5)^2)\ P \times R}{0.25\ (P+R)}$$

$\beta = 2$ (F2-score)
→ when FN is more important than FP

$$= \frac{(1+4)\ P \times R}{4\ (P+R)} = \frac{5\ (P \times R)}{4\ (P+R)}$$